

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper: CS-104b — Computer Org & Arch

Time: Three hours

Maximum Marks: 70

Instructions: Answer five questions in all including Question No. 1, which is compulsory. Answers should be brief and to the point and may be supplemented with neat sketches, wherever required.

Question 1.

- (a) A cache memory has 8 cache lines (0-7). The memory block requests are in the order: 4, 3, 25, 8, 19, 6, 25, 8, 16, 35, 45, 22, 8, 3, 16, 25, 7.
- (i) Which cache line will have memory block 7 if fully associative mapping with LRU policy is used? Also, calculate the hit ratio and miss ratio.
- (ii) Which cache line will have memory block 7 if direct mapping is used? Also, calculate the hit ratio and miss ratio.

[7]

- (b) What are the basic differences among a branch instruction, a call subroutine instruction, and a program interrupt?

[7]

Question 2.

- (a) Design a 4-bit Binary Adder-Subtractor and a Binary Incrementor. Explain their working.
- [6]
- (b) Why is register transfer language preferred for describing the internal organization of digital computers?
- [4]
- (c) What is an instruction cycle, and what are the phases of the instruction cycle?
- [4]

Question 3.

- (a) Design a bus system using three-state buffers. The number of registers is 6, and the size of each register is 4 bits.
- [6]
- (b) What is the difference between isolated I/O and memory-mapped I/O?
- [4]
- (c) Write the differences between Static RAM and Dynamic RAM.
- [4]

Question 4. A computer uses a memory unit with 1 Giga words of 16 bits each. The computer supports three addressing modes (register mode, register indirect mode, direct mode) and 35 unique micro-operations. The instruction has four parts: an addressing mode part, an operation code, a register code part to specify one of 180 registers, and an address part.

- (a) (i) Find the minimum number of words required for storing one instruction.
(ii) Draw the instruction word format and indicate the number of bits in each part. [6]
- (b) Explain any four addressing modes with suitable examples. [4]
- (c) Explain any four applications of the stack in computer organization. [4]

Question 5.

- (a) What is virtual memory? Explain Virtual address Mapping using Pages with necessary examples. [7]
- (b) What is the locality of reference in cache memory? Explain its major types and how they are achieved. [4]
- (c) Explain word addressable and byte addressable memory. [3]

Question 6.

- (a) We have to provide a memory capacity of 4096 bytes using 128×8 RAM chips.
 - (i) How many chips are required?
 - (ii) How many lines of the address bus must be used to access 4096 bytes of memory?
 - (iii) How many of lines in (ii) will be common to all chips?
 - (iv) How many lines must be decoded for chip select? Specify the size of decoders. [5]
- (b) What is an interrupt? Explain its major types. [5]
- (c) List out the importance of the I/O interface. [4]

Question 7.

- (a) Explain the working of DMA (Direct Memory Access) with the block diagram. [8]
- (b) What is the disadvantage of the strobe method? Explain how the handshake method solves the problem. [6]